

The Tangential Hinge Field

A Complete Coordinate Arrangement, Entry, Addressing, and Alignment Specification

Canonical status

Defined structural model. This document specifies geometry and invariants only. Execution, storage, transport, and application semantics are external profiles.

Version 1.0 • 27 July 2026

Abstract

The Tangential Hinge Field defines a mirrored coordinate geometry centered on a dual zero hinge. The negative-zero coordinate P0 supplies the FIELD parity reference; the positive-zero coordinate T0 supplies the FIELD tangential source. Four mirrored plane classes—capacity, order, weight, and scalar—are arranged around the hinge. The field is addressed through a 16×16 Blackboard and entered through four ordered scope coordinates: FS, GS, RS, and US.

The decisive simplification of this version is that the scope coordinates replace independent definition planes as the only required entry structure. FS, GS, and RS are ordered descriptions of one completed US field. The outer planes are fixed references by position; they do not require separate entry words and cannot independently redefine the source.

This specification is self-contained as geometry. It defines arrangement, hinge authority, entry, addressing, mirrored alignment, and nested adjudication scales. It intentionally does not define runtime continuation, mutation, storage layout, transport, or application meaning.

Specification Status

Status	Meaning in this document
Defined model	A precise structural coordinate definition and a candidate proof target.
Arithmetic identity	An exact finite equality, count, or radix relation.
Invariant	A condition every conforming representation must preserve.
External profile	An optional OMI, runtime, storage, or transport binding outside the core geometry.

1. Canonical Lock

```
FIELD = [S- W- O- C- | P0:T0 | C+ O+ W+ S+]
```

```
HINGE    = P0:T0
```

```
ENTRY    = {FS, GS, RS, US}
```

```
ADDRESS  = 0xRC
```

```
REFERENCE = (ENTRY, ADDRESS, P0:T0)
```

A complete field reference requires only a scope entry, a Blackboard address, and the parity-source hinge. Capacity, order, weight, and scalar do not appear in the entry signature because their places are already fixed by the FIELD arrangement.

Canonical statement

The Tangential Hinge Field is a mirrored positional geometry whose only entry coordinates are FS,

GS, RS, and US, whose only definition authority is the zero hinge P0:T0, and whose complete reference is (ENTRY, ADDRESS, P0:T0).

2. Scope and Boundary of the Specification

The core specification defines:

- the mirrored plane arrangement;
- the parity-source hinge;
- the FS–GS–RS–US entry hierarchy;
- the 16×16 Blackboard address space;
- local/remote mirror projection;
- pairwise alignment conditions;
- Metatron and Tetragrammatron adjudication scales;
- the invariants required of a conforming representation.

The core specification does not define:

- instruction execution or continuation;
- mutation, assignment, or a state machine;
- file formats or byte serialization;
- network transfer or distributed consensus;
- application meanings for capacity, order, weight, or scalar;
- an implementation language.

Older runtime words such as META, ESC, NEXT, PORT, CAR, CDR, and CONS may appear in external OMI profiles or historical documents, but they are not required to define this field geometry.

3. Structural Vocabulary

Term	Structural definition
NULL	An unbound coordinate place. NULL is explicit; it is not hidden nil.
BIND	A directly established coordinate place.
FS	The least-direct scope description: one direct place and three subordinate references.
GS	A scope description with two direct places and two subordinate references.
RS	A scope description with three direct places and one subordinate reference.

Term	Structural definition
US	The completed scope description: four directly established places.
P0	Negative-zero FIELD parity reference.
T0	Positive-zero FIELD tangential source.
C	Capacity plane class, magnitude 1.
O	Order plane class, magnitude 2. The letter O must not be confused with the numeral 0.
W	Weight plane class, magnitude 3.
S	Scalar plane class, magnitude 4.

Notation lock

`O` means order. `0` means zero. Therefore `P0` and `T0` are zero-oriented hinge coordinates; `O−` and `O+` are order-plane coordinates.

4. Field Arrangement

Negative arrangement				Hinge	Positive arrangement			
S−	W−	O−	C−	P0:T0	C+	O+	W+	S+
-4	-3	-2	-1	-0:+0	+1	+2	+3	+4

The signs identify orientation around the zero hinge. They do not inherently mean false/true, past/future, input/output, or lesser/greater. Any such reading belongs to an external profile.

Magnitude	Negative orientation	Positive orientation	Plane class
0	P0	T0	Parity reference / tangential source
1	C−	C+	Capacity
2	O−	O+	Order
3	W−	W+	Weight
4	S−	S+	Scalar

The field is palindromic at the class level: the negative arrangement approaches zero in the order scalar-weight-order-capacity, and the positive arrangement departs zero in the inverse order capacity-order-weight-scalar.

5. The P0:T0 Tangent Hinge

$$Q^- = [S^- \ W^- \ O^- \ C^-]$$

$$Q^+ = [C^+ \ O^+ \ W^+ \ S^+]$$

$$\text{FIELD} = Q^- \mid P0:T0 \mid Q^+$$

P0:T0 is the tangent hinge: the exact central relation at which arrangement becomes alignment. T0 originates the tangential FIELD source. P0 supplies the FIELD parity reference against which the source-relative arrangement may be checked.

Definition-authority invariant

A FIELD definition may originate only at T0 and may be admitted only relative to P0. Outer planes are oriented references to the source and may not independently redefine it.

5.1 Width-agnostic zero

Coordinate width	Zero form
4-bit nibble	0x0
8-bit byte	0x00
16-bit word	0x0000
32-bit quadrant	0x00000000
64-bit block	0x0000000000000000

The literal expands to the declared width; the hinge relation does not change. Signed zero here is orientation, not distinct integer arithmetic: -0 names the parity-facing zero and $+0$ names the source-facing zero.

6. FS-GS-RS-US Entry Hierarchy

The prior definition-heavy form is replaced by a scope-entry model. FS, GS, and RS are not separate definitions beside US. They are ordered descriptions of one completed US field.

$$FS_1 := [BIND, GS_1, RS_1, US_1]$$

$$GS_1 := [BIND, RS_1, US_1, BIND]$$

$$RS_1 := [BIND, US_1, BIND, BIND]$$

$$US_1 := [BIND, BIND, BIND, BIND]$$

Square brackets are used deliberately to emphasize fixed places. These forms are not Lisp calls, stream instructions, or executable lists.

Entry	Direct BIND places	Subordinate references	Structural reading
FS	1	3: GS, RS, US	Outermost description of the completed field
GS	2	2: RS, US	More direct description
RS	3	1: US	Directly approaches the completed field
US	4	0	Completely bound four-place field

```
direct BIND places + subordinate reference places = 4
```

```
FS → GS → RS → US
```

The completed tangential field attaches to US:

```
US1.FIELD = [S- W- 0- C- | P0:T0 | C+ 0+ W+ S+]
```

The suffix identifies an instance or layer. It does not alter the four-place scope grammar.

7. Arrangement and Alignment

7.1 Arrangement

Arrangement assigns every coordinate its fixed magnitude, orientation, and place. It answers: where does each reference reside?

```
S- W- 0- C- | P0:T0 | C+ 0+ W+ S+
```

7.2 Alignment

Alignment compares each negative coordinate with its positive partner. In an equal-width binary profile, XOR exposes the difference without changing either reference:

```
ΔC = C- XOR C+
ΔO = 0- XOR 0+
ΔW = W- XOR W+
ΔS = S- XOR S+
```

Exact alignment:

```
ΔC = 0
ΔO = 0
ΔW = 0
```

$$\Delta S = 0$$

The four witnesses must remain independently available. A single combined XOR is not sufficient to prove complete alignment because distinct nonzero differences can cancel.

Arrangement-hinge-alignment law

Arrangement fixes the mirrored places. P0:T0 fixes the central relation. Alignment checks each mirrored pair.

8. The 16×16 Blackboard Address Surface

```
row    = high nibble
column = low nibble
address = 0xRC
```

$R \in 0x0 \dots 0xF$

$C \in 0x0 \dots 0xF$

The Blackboard contains $16 \times 16 = 256$ byte coordinates covering 0x00 through 0xFF. At every address, the same field schema is available:

```
FIELD[0xRC] =
[S- W- O- C- | P0:T0 | C+ O+ W+ S+]
```

8.1 Local and remote macro-surfaces

Surface	Address range	Selector condition
Local	0x00...0x7F	bit 7 = 0
Remote	0x80...0xFF	bit 7 = 1

$$\text{REMOTE} = \text{LOCAL} \text{ XOR } 0x80$$

The projection preserves the low seven address bits and changes only the macro-surface selector. Thus every local coordinate has one exact remote counterpart.

9. Canonical Bounding Rectangles

Lens	Corner coordinates
Local 0	(0x00, 0x07, 0x37, 0x30)
Local 1	(0x08, 0x0F, 0x3F, 0x38)
Local 2	(0x40, 0x47, 0x77, 0x70)

Lens	Corner coordinates
Local 3	(0x48, 0x4F, 0x7F, 0x78)
Remote 0	(0x80, 0x87, 0xB7, 0xB0)
Remote 1	(0x88, 0x8F, 0xBF, 0xB8)
Remote 2	(0xC0, 0xC7, 0xF7, 0xF0)
Remote 3	(0xC8, 0xCF, 0xFF, 0xF8)

Each rectangle contains 4 rows × 8 columns = 32 coordinates. Eight rectangles therefore cover the complete 256-coordinate Blackboard.

$$8 \text{ lenses} \times 32 \text{ coordinates} = 256 \text{ coordinates}$$

10. The 32-Position Tangential Lens

One 32-position lens supports several nested terminal views. The lens does not change; only the adjudication boundary changes.

Partition	Body	Terminal field	Structural view
24 + 8	24	8	Four-scope permutation body plus complete COBS-CONS terminal
26 + 6	26	6	Operational OMINO gauge
27 + 5	27	5	Tetragrammatron relation governor
28 + 4	28	4	Metatron incidence field

```

8-position field: [8 9 A B C D E F]
6-position field:   [A B C D E F]
5-position field:   [B C D E F]
4-position field:   [C D E F]
```

The 24-position body also matches the $4! = 24$ possible orderings of FS, GS, RS, and US. This establishes the local scope-order resolution without altering the coordinates themselves.

11. Tangential Gauge Rows

Gauge	Origins	Control completions
Low	0x20, 0x30, 0x40, 0x50, 0x60, 0x70	0x2F, 0x3F, 0x4F, 0x5F, 0x6F, 0x7F
High	0xA0, 0xB0, 0xC0, 0xD0, 0xE0, 0xF0	0xAF, 0xBF, 0xCF, 0xDF, 0xEF, 0xFF

```
row form = X0 ... XF
```

```
X0 = zero-relative tangential origin
```

```
XF = complete four-place tangential closure
```

The low affine gauge occupies 0x00...0x1F. The high projective gauge occupies 0x80...0x9F. The low/high relation is the same 0x80 mirror involution used by the Blackboard.

12. Nested Adjudication Scales

12.1 General four-place law

```
For radix B:
```

```
anchors = B0, B1, B2, B3, B4
```

```
domain = 0 ... B4 - 1
```

The first four anchors establish place weights. The fifth is the exclusive upper boundary of the four-place domain.

12.2 Metatron: radix-16 incidence

```
160 = 0x000001
```

```
161 = 0x000010
```

```
162 = 0x000100
```

```
163 = 0x001000
```

```
164 = 0x100000
```

```
Metatron domain = 164 = 65,536 = 216
```

```
admissible values = 0x0000 ... 0xFFFF
```

Metatron validates the internal place structure of one four-nibble, 16-bit field.

12.3 Tetragrammatron: radix-256 relation

```
2560 = 0x0000000001
```

```
2561 = 0x00000000100
```

```
2562 = 0x00000100000
```

```
2563 = 0x00010000000
```

```
2564 = 0x10000000000
```

```
Tetragrammatron domain =  $256^4 = 4,294,967,296 = 2^{32}$ 
admissible values = 0x00000000 ... 0xFFFFFFFF
```

Tetragrammatron validates the four-byte or four-page relation. The nine-hex-digit terminal anchor 0x100000000 is intentional: it is the exclusive upper boundary above the complete 32-bit domain.

Nested law

Metatron checks four hexadecimal places inside a 16-bit word. Tetragrammatron checks four byte-scale places inside a 32-bit quadrant. Both instantiate the same B^4 law at different radices.

13. Gnomonic Projection Azimuth

```
GNOMONIC PROJECTION AZIMUTH = 0xAA55
```

```
0x2 XOR 0x8 = 0xA
```

```
0x8 XOR 0xA = 0x2
```

```
0xA XOR 0x2 = 0x8
```

```
0xA XOR 0xF = 0x5
```

The value 0xAA55 supports both nibble and byte readings:

```
nibble view: A | A | 5 | 5
```

```
byte view:  AA | 55
```

In the broader OMI architecture, this coordinate acts as the orientation bridge between Metatron nibble-scale incidence and Tetragrammatron byte-scale relation. In this agnostic specification, it is an optional external orientation profile rather than a required component of the FIELD lock.

14. Complete Reference Model

```
ENTRY    ∈ {FS, GS, RS, US}
```

```
ADDRESS  = 0xRC
```

```
HINGE    = P0:T0
```

```
REFERENCE = (ENTRY, ADDRESS, P0:T0)
```

Component	Question answered
ENTRY	Through which ordered description of US is the field approached?
ADDRESS	At which Blackboard coordinate is the field referenced?
P0:T0	What is the central parity-source relation of the field?

```

FS / GS / RS / US
  → FIELD[0xRC]
  → P0:T0

```

The mirrored outer planes are implicit in FIELD[0xRC]. They are not additional arguments.

15. Conformance Invariants

Invariant	Required condition
Field form	Exactly [S− W− O− C− P0:T0 C+ O+ W+ S+]
Order/zero notation	O denotes order; 0 denotes zero.
Scope width	Every scope description has exactly four places.
Scope progression	Direct BIND count is 1, 2, 3, 4 for FS, GS, RS, US.
Scope invariant	Direct places + subordinate references = 4.
Definition authority	Only T0 originates the source; P0 supplies its parity reference.
Mirror pairing	$C \leftrightarrow C^+$, $O \leftrightarrow O^+$, $W \leftrightarrow W^+$, $S \leftrightarrow S^+$.
Exact alignment	$\Delta C = \Delta O = \Delta W = \Delta S = 0$.
Address domain	Blackboard addresses cover exactly 0x00...0xFF.
Macro-surface mirror	REMOTE = LOCAL XOR 0x80.
Bounding coverage	$8 \times 32 = 256$ Blackboard coordinates.
Metatron domain	16^4 values with five anchors $16^0 \dots 16^4$.
Tetragrammatron domain	256^4 values with five anchors $256^0 \dots 256^4$.

16. Minimum Conformance Vectors

A representation of the specification should at minimum verify:

- all four scope rows have exactly four places;
- FS, GS, RS, and US have direct BIND counts 1, 2, 3, and 4;
- every subordinate reference moves toward US and no form refers upward;
- the FIELD has ten oriented positions over five magnitudes;
- P0 and T0 occupy negative and positive zero respectively;
- local addresses map to remote addresses by XOR 0x80 and round-trip under the same operation;

- the eight canonical rectangles are non-overlapping and cover all 256 byte coordinates;
- each 32-position lens satisfies 24+8, 26+6, 27+5, and 28+4;
- Metatron anchors equal 16^0 through 16^4 ;
- Tetragrammatron anchors equal 256^0 through 256^4 ;
- alignment is accepted only when all four paired deltas independently equal zero.

17. Agnosticism and Interpretation Boundary

Capacity, order, weight, and scalar are coordinate-class names. They do not prescribe physical units, business semantics, programming types, or human meanings. A domain profile may interpret them, but the profile may not alter their positions, orientations, or pairing.

Likewise, negative and positive orientation do not prescribe chronology or polarity. P0 and T0 do not assert metaphysical, logical, or physical truth. They name the parity-facing and source-facing sides of the dual zero hinge.

Agnostic completeness

The model is complete because every structural role required for arrangement, entry, addressing, and alignment is defined. Application meaning is intentionally outside its jurisdiction.

18. Optional OMI Architecture Bindings

The following bindings document how the geometry may connect to the larger OMI architecture. They are not part of the agnostic core.

Optional term	Possible architecture binding
COBS-CONS	Blackboard framing or terminal gauge profile
CAR / CDR	Local / remote ordered relation roles
*.omi / *.imo	User-local / user-remote representation surfaces
CONS = 0xFFFFFFFF	Complete four-byte closure target
0xAA55	Gnomonic orientation profile
OMINO / OMICRON / OMNICON	Six-, seven-, and eight-position operational/integrity profiles
Algorithmic Clock / Slide Ruler	External synchrony and rotation coordinates

These bindings may use the FIELD, but they may not redefine the FIELD lock.

19. Migration from the Earlier Definition-Heavy Form

The historical draft combined the field geometry with a runtime metalanguage containing XOR, META, ESC, and five-place forms such as `(BIND GS RS US ESC)`. That material remains meaningful in its own runtime specification, but it obscured the self-contained geometry by implying continuation and cancellation semantics.

Earlier draft	Canonical structural replacement
Five-place scope row ending in ESC	Four-place scope description only
BIND as an operation to be implemented	BIND as a directly established coordinate place
ESC/META inside the entry hierarchy	Moved to an external runtime profile
Parentheses suggesting Lisp evaluation	Square brackets indicating fixed places
Each plane approached as an independent definition	FS/GS/RS describe one completed US field
State as a central data category	T0 is tangential source; P0 is parity reference

Supersession rule

For the Tangential Hinge Field itself, the four-place scope forms in this document supersede earlier five-place forms containing ESC or NEXT. Earlier forms may remain valid only within a separately named runtime profile.

20. Canonical Summary

```
FIELD = [S- W- 0- C- | P0:T0 | C+ 0+ W+ S+]
```

```
P0 = negative-zero FIELD parity reference
```

```
T0 = positive-zero FIELD tangential source
```

```
FS1 := [BIND, GS1, RS1, US1]
```

```
GS1 := [BIND, RS1, US1, BIND]
```

```
RS1 := [BIND, US1, BIND, BIND]
```

```
US1 := [BIND, BIND, BIND, BIND]
```

```
US1.FIELD = FIELD
```

```
ENTRY    = {FS, GS, RS, US}
```

```
ADDRESS  = 0xRC
```

```
HINGE    = P0:T0
```

```
REFERENCE = (ENTRY, ADDRESS, HINGE)
```

```
ΔC = C- XOR C+
```

```
Δ0 = 0- XOR 0+
```

$$\Delta W = W^- \text{ XOR } W^+$$

$$\Delta S = S^- \text{ XOR } S^+$$

$$\text{ALIGNMENT} \Leftrightarrow \Delta C = \Delta O = \Delta W = \Delta S = 0$$

$$\text{REMOTE} = \text{LOCAL} \text{ XOR } 0x80$$

Final canonical statement

The Tangential Hinge Field is a complete and well-defined structural coordinate specification. It defines a mirrored capacity-order-weight-scalar arrangement centered on the P0:T0 parity-source hinge; four ordered scope entries that describe one completed US field; a complete 16×16 Blackboard address surface; exact local/remote mirror projection; pairwise alignment; and nested Metatron and Tetragrammatron adjudication scales. It is geometry, not an execution language.

Appendix A. Formal Structural Grammar

```

<field>      ::= "[" <negative> "|" <hinge> "|" <positive> "]"
<negative>   ::= S- W- O- C-
<hinge>     ::= P0 ":" T0
<positive>   ::= C+ O+ W+ S+

<entry-name> ::= FS | GS | RS | US
<place>      ::= BIND | <entry-name>

<FS>        ::= [BIND, GS, RS, US]
<GS>        ::= [BIND, RS, US, BIND]
<RS>        ::= [BIND, US, BIND, BIND]
<US>        ::= [BIND, BIND, BIND, BIND]

<address>    ::= 0xRC
<reference>  ::= "(" <entry-name> "," <address> "," P0:T0 ")"

```

Appendix B. Blackboard Bounding Matrix

```

Local:
  (0x00, 0x07, 0x37, 0x30)
  (0x08, 0x0F, 0x3F, 0x38)
  (0x40, 0x47, 0x77, 0x70)
  (0x48, 0x4F, 0x7F, 0x78)

Remote:
  (0x80, 0x87, 0xB7, 0xB0)
  (0x88, 0x8F, 0xBF, 0xB8)
  (0xC0, 0xC7, 0xF7, 0xF0)
  (0xC8, 0xCF, 0xFF, 0xF8)

```

Appendix C. Adjudication Anchor Tables

Exponent	Metatron: 16^n	Tetragrammatron: 256^n
0	0x00001	0x000000001
1	0x00010	0x000000100
2	0x00100	0x000010000
3	0x01000	0x001000000
4	0x10000	0x100000000

Appendix D. Historical Source Map

This specification consolidates and narrows concepts that appeared across prior OMI documents. The historical material established the 16×16 positional lens, the 0x80 low/high mirror, ordered scope coordinates, nested Metatron/Tetragrammatron scales, and the distinction between complete coordinates and reduced witnesses. This document isolates the geometry from the runtime.

Historical contribution	Treatment here
16×16 hexadecimal coordinate lens	Retained as the complete Blackboard address surface.
Low/high projection by XOR 0x80	Retained as the local/remote mirror invariant.
FS > GS > RS > US	Retained and normalized as four-place scope descriptions.
Five-place forms ending in ESC	Excluded from the core; reserved for a runtime profile.
Metatron and Tetragrammatron	Retained as nested radix-16 and radix-256 adjudication scales.
0xAA55 azimuth	Retained as an optional external orientation profile.
CONS/CAR/CDR and serialization	Classified as optional architecture bindings.
4320 clock and 5040 ruler	Excluded from this geometry; remain separate timing coordinates.